Inequalities Lecture Scribe

L25_Problem_Solving__Tutorial (1)(1).pdf
PDF

Generate a problem-oriented lecture scribe for L25 – Tutorial Session (Inequalities) of CSE400 – Fundamentals of Probability in Computing, based on the provided problem sheet.

The scribe must be structured according to the questions in the tutorial, and should serve as exam-focused preparation material, helping a student understand how to solve such problems.

Here's a **problem-oriented, exam-focused lecture scribe** for *L25 – Tutorial Session (Inequalities)* based directly on the given problem sheet. It is structured **question-wise**, with formulas, reasoning, and solving approach emphasized.

📘 Lecture Scribe: Inequalities (CSE400 – L25)

This session focuses on **Markov, Chebyshev, Jensen, and Chernoff inequalities**, and how to apply them to bound probabilities and expectations.

🔹 Q1: Markov's Inequality

Key Concept:
For a non-negative random variable X,

$$P(X \geq a) \leq \frac{E[X]}{a}$$

(a) Income ≥ 120,000

- Given: $E[X] = 40000$, $a = 120000$

$$P(X \geq 120000) \leq \frac{40000}{120000} = \frac{1}{3}$$

✅ **Final Answer:** Upper bound = **1/3**

(b) Income < 10,000?

- Markov only works for $P(X \geq a)$
- Cannot directly bound $P(X < a)$

❌ **Conclusion:** Not applicable

🔹 Q2: Chebyshev's Inequality

Key Concept:

$$P(|X-\mu| \geq k\sigma) \leq \frac{1}{k^2}$$

(a) Defective resistor

- Mean = 100, range = [94, 106]
- Deviation = 6 $\rightarrow k = \frac{6}{2} = 3$

$$P(|X-100| \geq 6) \leq \frac{1}{3^2} = \frac{1}{9}$$

✅ **Upper bound = 1/9**

(b) Out of 10,000 resistors

$$\text{Max defective} = 10000 \cdot \frac{1}{9} \approx 1111$$

🔹 Q3: Convexity (Exponential Function)

Goal:

$$E[e^{tY}] \leq 1 - \mu + \mu e^t$$

Idea:

- e^x is convex
- Use line joining points at 0 and 1:

$$e^{tY} \leq (1-Y)e^0 + Ye^t$$

Take expectation:

$$E[e^{tY}] \leq 1 - \mu + \mu e^t$$

✅ **Technique:** Convexity + linear bound

🔹 Q4: Jensen's Inequality

Key Concept:
For convex g,

$$E[g(X)] \geq g(E[X])$$

Apply for $g(X) = X^a, a > 1$:

$$E[X^a] \geq (E[X])^a$$

✅ **Important:** Power functions with $a > 1$ are convex

🔹 Q5: Standard Chebyshev Form

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Derivation idea:

- Use variance definition
- Substitute $a = k\sigma$

✅ **Key takeaway:**
Deviation ≥ $k\sigma$ is very unlikely for large k

🔹 Q6: Production Example

(a) $P(X \geq 1000)$

Using Markov:

$$P(X \geq 1000) \leq \frac{500}{1000} = \frac{1}{2}$$

(b) $P(400 \leq X \leq 600)$

- Mean = 500, variance = 100 → $\sigma = 10$
- Range = ±100 → $k = 10$

$$P(|X - 500| \geq 100) \leq \frac{1}{10^2} = \frac{1}{100}$$

$$P(400 \leq X \leq 600) \geq 1 - \frac{1}{100} = \frac{99}{100}$$

🔹 Q7: Chernoff Bound Setup

(a) For $X_i \in \{-1, 1\}$

$$E[e^{tX_i}] = \frac{e^t + e^{-t}}{2} \leq e^{t^2/2}$$

(b) Sum S_n

Independence ⇒ multiply expectations:

$$E[e^{tS_n}] = \prod E[e^{tX_i}] \leq (e^{t^2/2})^n = e^{nt^2/2}$$

✅ **Important:** Moment generating function (MGF)

🔹 Q8: Pairing Problem

- Total: 100 pairs
- Want ≤ 30 mixed pairs

Approach:

- Expected mixed pairs ≈ 50
- Use **Chernoff bound** to bound large deviation

General form:

$$P(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}$$

Here:

- $\mu = 50$, target = 30 ⇒ $\delta = 0.4$

$$P(X \leq 30) \leq e^{-50(0.4)^2/2} = e^{-4}$$

🔹 Q9: Inequality Proof

$$\sum_{i=1}^{n} \sqrt{i^2+1} \geq \frac{n}{2}\sqrt{n^2+2n+5}$$

Idea:

- Use bounding + symmetry
- Compare with integral or midpoint method

✅ **Exam Tip:** Use pairing or convexity arguments

🔹 Q10: Poisson Bounds

Let $X \sim \text{Poisson}(20)$

(a) Markov

$$P(X \geq 26) \leq \frac{20}{26}$$

(b) One-sided Chebyshev

$$P(X - 20 \geq 6) \leq \frac{20}{6^2} = \frac{20}{36}$$

(c) Chernoff Bound

General:

$$P(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

Here:

- $\mu = 20, \delta = 0.3$

$$P(X \geq 26) \leq \left(\frac{e^{0.3}}{1.3^{1.3}}\right)^{20}$$

✅ **Best bound among all methods**

🧠 Final Exam Takeaways

📌 When to Use What:

Inequality	Use Case
Markov	Only mean known
Chebyshev	Mean + variance
Jensen	Convex/concave functions
Chernoff	Strong exponential bounds

📌 Key Strategy:

- Identify **given information (mean, variance, distribution)**
- Choose **tightest inequality**
- Express in **standard form**
- Compute **k or δ correctly**

If you want, I can convert this into **last-day revision notes or cheat sheet (1–2 pages)** for quick exam prep.

L25_Problem_Solving__Tutorial (1)(2).pdf
PDF

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Here is a **clean, problem-oriented lecture scribe** for **L25 – Tutorial Session (Inequalities)**, structured exactly according to the tutorial sheet and designed for **exam preparation and problem-solving clarity**.
L25_Problem_Solving__Tutorial (…

📘 L25 Scribe: Inequalities (Problem-Oriented)

This tutorial focuses on **how to apply inequalities to bound probabilities and expectations** when full distributions are unknown.

🔹 Q1: Markov's Inequality

🔑 Concept

Use when:

- Random variable is **non-negative**
- Only **expectation is known**

$P(X \geq a) \leq \frac{E[X]}{a}$

(a) Income ≥ 120,000

Steps:

1. Identify:
   - $E[X] = 40000$
   - $a = 120000$
2. Apply formula:

$$P(X \geq 120000) \leq \frac{40000}{120000} = \frac{1}{3}$$

✅ **Answer:** $\boxed{1/3}$

(b) Income < 10,000

Key Insight:

- Markov only bounds **upper tail**
- Cannot directly handle $P(X < a)$

❌ **Conclusion:** Not applicable

🔹 Q2: Chebyshev's Inequality

🔑 Concept

Use when:

- Mean + variance known
- Want deviation probability

$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

(a) Defective resistor

Steps:

1. Mean = 100, σ = 2
2. Range = [94,106] → deviation = 6
3. Compute:

$$k = \frac{6}{2} = 3$$

4. Apply:

$$P(\text{defective}) \leq \frac{1}{9}$$

✅ **Answer:** $\boxed{1/9}$

(b) Expected defective (10,000 items)

$$10000 \times \frac{1}{9} \approx 1111$$

✅ **Answer:** Maximum ≈ **1111 defective**

🔹 Q3: Convexity (Exponential Bound)

🔑 Idea

- e^x is convex
- Use line between endpoints (0 and 1)

Key step:

$$e^{tY} \leq (1-Y)e^0 + Ye^t$$

Take expectation:

$$E[e^{tY}] \leq 1 - \mu + \mu e^t$$

✅ **Technique:** Convexity + linear upper bound

🔹 Q4: Jensen's Inequality

🔑 Concept

For convex g:

$$E[g(X)] \geq g(E[X])$$

Apply for $g(X) = X^a$, $a > 1$:

$$E[X^a] \geq (E[X])^a$$

✅ **Interpretation:** Variability increases cost

🔹 Q5: Standard Chebyshev Form

Goal:

Show:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Method:

- Start with Chebyshev general form
- Substitute $a = k\sigma$

✅ **Conclusion:** Larger deviations → rapidly decreasing probability

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🔹 Q6: Production Example

(a) $P(X \geq 1000)$

Using Markov:

$$P(X \geq 1000) \leq \frac{500}{1000} = \frac{1}{2}$$

(b) $P(400 \leq X \leq 600)$

Steps:

1. Mean = 500, variance = 100 → σ = 10
2. Range = ±100 → $k = 10$

$$P(|X - 500| \geq 100) \leq \frac{1}{100}$$

$$P(400 \leq X \leq 600) \geq 1 - \frac{1}{100} = \frac{99}{100}$$

🔹 Q7: Chernoff Bound (MGF Method)

(a) Single variable

$$E[e^{tX_i}] = \frac{e^t + e^{-t}}{2} \leq e^{t^2/2}$$

(b) Sum S_n

Using independence:

$$E[e^{tS_n}] = \prod E[e^{tX_i}] \leq e^{nt^2/2}$$

✅ **Core idea:** Exponential bounding via MGF

🔹 Q8: Pairing Problem

Setup:

- Expected mixed pairs ≈ 50
- Want ≤ 30 → large deviation

Use Chernoff:

$$P(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}$$

- $\mu = 50$
- $\delta = 0.4$

$$P(X \leq 30) \leq e^{-4}$$

✅ **Very small probability**

🔹 Q9: Inequality Proof

$$\sum_{i=1}^{n} \sqrt{i^2+1} \geq \frac{n}{2}\sqrt{n^2+2n+5}$$

Approach:

- Use **pairing / bounding**
- Compare terms symmetrically
- Or apply convexity / integral bounds

✅ **Exam Tip:** Don't expand—use structure

🔹 Q10: Poisson Bounds

Given $X \sim \text{Poisson}(20)$

(a) Markov

$$P(X \geq 26) \leq \frac{20}{26}$$

(b) One-sided Chebyshev

$$P(X - 20 \geq 6) \leq \frac{20}{36}$$

(c) Chernoff (tightest)

$$P(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

- $\delta = 0.3$

🧠 Final Exam Strategy

🔥 Which inequality to use?

Given Info	Use
Only mean	Markov
Mean + variance	Chebyshev
Convex function	Jensen
Strong exponential bounds	Chernoff

⚡ Problem-Solving Flow

1. Identify known values (mean, variance, bounds)
2. Match inequality type
3. Convert to standard form
4. Solve carefully for k, a, or δ

🚀 Key Insight

- **Markov → weak bound**
- **Chebyshev → moderate**
- **Chernoff → strongest**

If you want, I can also turn this into:

- **1-page cheat sheet**
- **important exam questions + patterns**
- **step-by-step solved examples like exam answers**

Sources

L25_Problem_Solving__Tutorial (1)(3).pdf
PDF

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Here is a **fully structured, problem-oriented lecture scribe** for **L25 – Tutorial Session (Inequalities)**, designed specifically for **exam preparation and problem-solving clarity**, strictly following the tutorial questions.
L25_Problem_Solving__Tutorial (…

📘 L25 Lecture Scribe: Inequalities (Exam-Focused)

This tutorial trains you to **bound probabilities using limited information** (mean, variance, or distribution type). The core tools are:

- Markov Inequality
- Chebyshev Inequality
- Jensen's Inequality
- Chernoff Bounds

🔹 Q1: Markov's Inequality (Basic Bounding)

🔑 When to use:

- Only **expectation is known**
- Random variable is **non-negative**

$P(X \geq a) \leq \frac{E[X]}{a}$

(a) $P(X \geq 120000)$

Steps:

- $E[X] = 40000$, $a = 120000$

$$P(X \geq 120000) \leq \frac{40000}{120000} = \frac{1}{3}$$

✅ **Final Answer:** $\boxed{1/3}$

(b) $P(X < 10000)$

Key Concept:

- Markov bounds only **upper tail probabilities**
- Cannot directly compute $P(X < a)$

❌ **Answer:** Not applicable

🔹 Q2: Chebyshev's Inequality (Variance-Based Bound)

🔑 When to use:

- Mean + variance given
- Want probability of deviation

$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$

(a) Defective resistor

Steps:

- Mean = 100, σ = 2
- Defective if outside [94, 106] → deviation = 6

$$k = \frac{6}{2} = 3$$

$$P(\text{defective}) \leq \frac{1}{9}$$

✅ **Answer:** $\boxed{1/9}$

(b) Maximum defective (10,000 items)

$$10000 \times \frac{1}{9} \approx 1111$$

✅ **Answer: ≈ 1111 resistors**

🔹 Q3: Convexity Inequality (Exponential)

🔑 Idea:

- e^x is convex
- Use linear upper bound between endpoints

Key step:

$$e^{tY} \leq (1-Y)e^0 + Ye^t$$

Taking expectation:

$$E[e^{tY}] \leq 1 - \mu + \mu e^t$$

✅ **Technique:** Convexity + expectation

🔹 Q4: Jensen's Inequality (Expectation Transformation)

🔑 Concept:

For convex $g(x)$,

$$E[g(X)] \geq g(E[X])$$

Apply for $g(X) = X^a, a > 1$:

$$E[X^a] \geq (E[X])^a$$

✅ **Interpretation:** Convex cost → average cost increases

🔹 Q5: Standard Chebyshev Form (Derivation)

Goal:

$$P(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}$$

Method:

- Start from:

$$P(|X-\mu| \geq a) \leq \frac{\sigma^2}{a^2}$$

- Substitute $a = k\sigma$

✅ **Conclusion:** Probability drops as $1/k^2$

🔹 Q6: Production Problem

(a) $P(X \geq 1000)$

Using Markov:

$$P(X \geq 1000) \leq \frac{500}{1000} = \frac{1}{2}$$

(b) $P(400 \leq X \leq 600)$

Steps:

- Mean = 500, variance = 100 → σ = 10
- Range = ±100 → $k = 10$

$$P(|X-500| \geq 100) \leq \frac{1}{100}$$

$$P(400 \leq X \leq 600) \geq 1 - \frac{1}{100} = \frac{99}{100}$$

🔹 Q7: Chernoff Bound (MGF Technique)

(a) Single variable

$$E[e^{tX_i}] = \frac{e^t + e^{-t}}{2} \leq e^{t^2/2}$$

(b) Sum S_n

Using independence:

$$E[e^{tS_n}] = \prod E[e^{tX_i}] \leq e^{nt^2/2}$$

✅ **Key Insight:** MGF simplifies exponential bounds

🔹 Q8: Pairing Problem (Large Deviation)

Setup:

- Expected mixed pairs = 50
- Want ≤ 30

Use Chernoff:

$$P(X \leq (1-\delta)\mu) \leq e^{-\mu\delta^2/2}$$

- $\delta = 0.4$

$$P(X \leq 30) \leq e^{-4}$$

✅ **Conclusion:** Extremely small probability

🔹 Q9: Inequality Proof

$$\sum_{i=1}^{n} \sqrt{i^2+1} \geq \frac{n}{2}\sqrt{n^2+2n+5}$$

Approach:

- Use **pairing technique** or
- Compare symmetric terms
- Or use **convexity / bounding**

✅ **Exam Tip:** Avoid brute-force expansion

🔹 Q10: Comparing Bounds (Markov vs Chebyshev vs Chernoff)

Given $X \sim \text{Poisson}(20)$

(a) Markov

$$P(X \geq 26) \leq \frac{20}{26}$$

(b) One-sided Chebyshev

$$P(X - 20 \geq 6) \leq \frac{20}{36}$$

(c) Chernoff (best bound)

$$P(X \geq (1+\delta)\mu) \leq \left(\frac{e^{\delta}}{(1+\delta)^{1+\delta}}\right)^{\mu}$$

- $\delta = 0.3$

✅ **Most accurate upper bound**

🧠 Final Exam Strategy

🔥 Inequality Selection

Situation	Use
Only mean known	Markov
Mean + variance	Chebyshev
Convex function	Jensen
Tight exponential bound	Chernoff

⚡ Problem-Solving Template

1. Identify given data
2. Choose correct inequality
3. Convert into standard form
4. Compute parameters carefully
5. Interpret bound

🚀 Key Insights

- Markov → weakest bound
- Chebyshev → moderate

- Chernoff → strongest
- Jensen → transforms expectations

If you want next step, I can give:

- **Last-day revision notes (1-page)**
- **Important exam patterns from this topic**
- **Step-by-step solved answers in exam writing format**

Sources